The paleomagnetism of the Michikamau anorthositic intrusion, Labrador

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Twenty-nine oriented cores have been collected from six sites in the Michikamau Intrusion in Labrador. This had previously been well dated at 1400 m.y. by the potassium–argon method. A fold test has been applied to the paleomagnetic results after alternating magnetic field demagnetization. The results suggest that the magnetization was acquired with the rocks in their present attitudes. The stability of the magnetization has been well tested and it is concluded that the observed magnetization is thermoremanent and represents the direction of the earth's field at the intrusion at the time of its formation. The paleomagnetic pole position for the intrusion lies at 0.5° S, 144.7° W, with $A_{95} = 5.6°$. It is difficult to compare this result with most other Precambrian results from North America due to uncertainties in age equivalences. However, a comparison with recently published results from well-dated rocks from Colorado–Wyoming suggests there have been no large relative movements between this region and Labrador for the last 1400 million years.

Introduction

The Michikamau Intrusion is one of a belt of anorthositic masses that extends from the coast of Labrador, southwest through the Grenville (structural) Province of Ontario and Quebec (Stockwell 1963) into the Grenville of northern New York State. Within the Grenville Province these masses have been metamorphosed; however, the Michikamau Intrusion, along with several others, lies north of the Grenville Front and is essentially unaltered. For this reason it appeared to be one of the most suitable anorthositic bodies for paleomagnetic investigation. Furthermore, geochronologic control is available from previous K–Ar radiometric work that indicates that the intrusion has an age of 1400 ± 80 m.y. Rocks of this age suitable for paleomagnetic studies are uncommon within the Canadian Shield, so the Michikamau Intrusion provides an almost unique chance to obtain a reliable paleomagnetic pole position relative to this part of the shield for this period.

The paleomagnetism of the Allard Lake anorthosite, another mass from this belt of intrusions, has been studied recently by Hargraves and Burt (1967). This body lies within the Grenville Province and was metamorphosed during the Grenville orogeny. Hargraves and Burt concluded that the observed remanent magnetization of the Allard Lake rocks originated during events associated with the Grenville orogeny, which ended approximately 900 m.y. ago.

Geological Setting and Sampling

The Michikamau Intrusion has been described in detail by Emslie (1965), and the geological data of Fig. 1 and subsequent description is taken largely from his report. The intrusion centers on latitude 54° 28′ N, longitude 64° 00′ W and underlies an area of approximately 2000 km^2. Within a border group of fine-grained rock of olivine basalt composition, with coarser olivine gabbros and pyroxene troctolites, are major units of leucotroctolite, anorthosite, and leuco-gabbro, and a group of ferrous-rich rocks of dioritic, granodioritic, and syenitic compositions. All of these units are remarkably fresh and unaltered. Rocks surrounding the intrusion are predominantly paragneisses, quartzo-feldspathic gneisses, migmatites, granite, and syenite.

The Michikamau Intrusion contains internal planar structures of two types. One is defined by planar orientation of plagioclase, the other consists of rhythmic mineralogic layering. The two types of structure are sensibly parallel where

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they occur together. Emslie (1965) concluded that the lamination and layering were formed by accumulation of crystals settling from a liquid magma and that some of the planes may have had primary dips of about 30 degrees.

Along the west shore of Michikamau Lake the primary structures show consistent eastward dips of between 50 and 80 degrees. These are markedly steeper than elsewhere in the intrusion and suggest that the west lake shore area has been separated from the main mass by a fault, and tilted steeply to the east. The transgressive relationship of ferrogranodiorite, a later stage differentiate of the intrusion, toward anorthosite provides evidence that the anorthosite was sufficiently consolidated to fail by fracture while the overlying magma was still essentially liquid.

Three potassium–argon age determinations on biotite are available to date the intrusion (Emslie 1964). Country-rock paragneiss near the southeast margin gave an age of 1520 ± 85 m.y., while biotite from leucotroctolite of the intrusion gave an age of 1400 ± 80 m.y. Granite around Michikamats Lake, which Emslie (1965) assumed to be related to granite dikes cutting the intrusion, gave an age of 1360 ± 80 m.y. The 1400 m.y. age is thus bracketed by maximum and minimum ages of 1520 and 1360 m.y. Morse (1964) has reported a potassium–argon age determination of 1480 ± 50 m.y. on biotite from possibly rheomorphic granodiorite that cuts the Kiglapait anorthosite intrusion. This intrusion lies some 320 km northeast of Michikamau and like the Michikamau Intrusion was not involved in the Grenville orogeny. Morse (1964) has suggested that the potassium–argon data indicate age equivalence of the two intrusions and gives the approximate age of their primary crystallization.

In August 1967 one of us (W. F. F.) obtained a number of drill cores, 3.2 cm (1¼″) in diameter, from six sites whose locations are shown in Fig. 1. The cores were oriented by means of a solar compass similar to that described by Gough and Opdyke (1963). Values for the strike and dip of the primary layering at each collection site are listed in Table I. These values were obtained from Maps 31-1963 and 3-1964 of the Geological Survey of Canada. Site 1, on Fraser Lake, is in rock mapped by Emslie (1965) as olivine gabbro of the border group. The remaining five sites are from his anorthosite unit.

Two cylindrical specimens, 2.9 cm (1⅛″) in length, were cut from each core. Each core is considered to represent an independent sample and in the statistical analyses the results from the two specimens were combined and the mean result was taken to be representative of the sample.

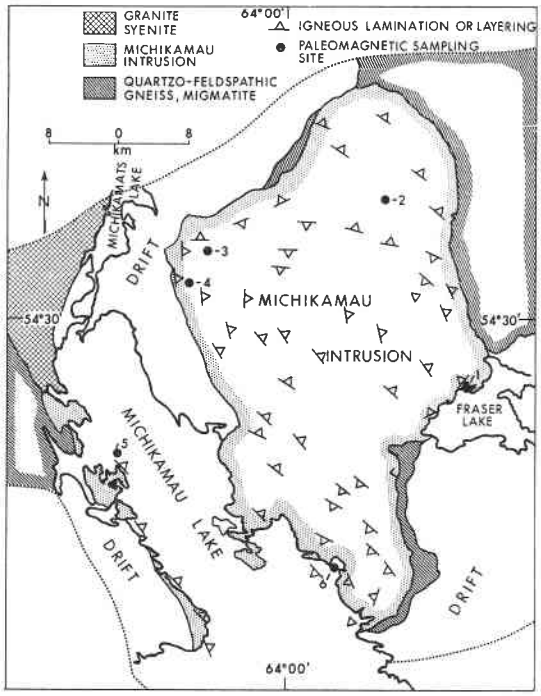


FIG. 1. Geologic map of the Michikamau Intrusion showing the paleomagnetic sampling sites.

Measurements

The natural remanent magnetization (NRM) of each specimen was measured by means of a spinner magnetometer (Princeton Applied Research Corporation, SM-1). The specimens were strongly magnetized with magnetic moments ranging from 6.6×10^{-6} to 2.7×10^{-4} A-m^2. The directions of NRM of the samples were well grouped at all sites except for site 3. However, even at this site the value of R, the length of the resultant of the N unit vectors, indicates that the directions of magnetization are not random on Watson's (1956) criterion. The parameters of the mean NRM direction are given in Table I for each of the six sites and the mean direction of NRM for each site is shown in Fig. 2. The means are smeared between the direction of the present earth's field at the Michikamau Intrusion and a direction of magnetization that is approximately horizontal and due west.

TABLE I

Site mean directions for the Michikamau anorthositic intrusion

Site	Igneous layering		N	M_N			B_{ac}	M_{ac}					Site Pole	
	Strike	Dip		D	I	k	mWb/m²	D	I	R	k	α_{95}	Lat.	Long.
1	34°	35°NW	6	266½	+41	36	30	256	+18½	5.945	91.1	6.0	0.1°S	136.8°W
2	126°	20°NE	5	253	+18½	14	30	262½	+6½	4.739	15.3	16	1.6°S	146.0°W
3	86°	30°SE	5	248½	+78½	4.8	30	259½	+22½	4.304	5.75	26	3.4°N	138.6°W
4	138°	25°NE	4	273	−2	37	20	267½	−2½	3.883	25.6	14	2.4°S	152.8°W
5	132°	67°NE	5	273	−1	11	30	262½	−9	4.958	94.7	6.4	8.0°S	150.7°W
6	130°	27°SW	4	263	+24	30	30	265	+21	3.968	95.0	7.2	5.8°N	143.5°W

NOTE: N = number of samples measured; R = length of the resultant of the N unit vectors; D = declination of remanent magnetization; I = inclination of remanent magnetization; M_N = NRM M_{ac} = RM after demagnetization in alternating field B_{ac}; $k = (N-1)/(N-R)$ = estimate of Fisher's κ parameter; α_{95} = semi angle of cone of 95% confidence for mean direction.

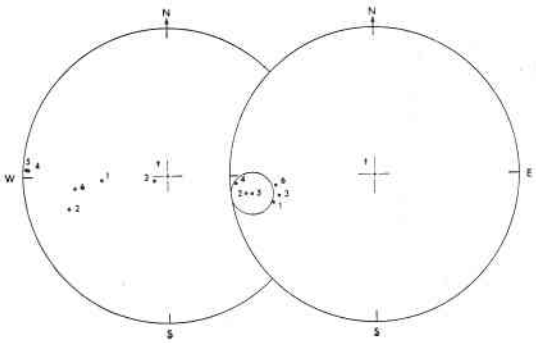


FIG. 2. Stereographic projection of the directions of magnetization of the six sites in the Michikamau Intrusion. Left-hand side, NRM directions. Right-hand side, directions of remanent magnetization after alternating field demagnetization. Solid-line circle shows 95% confidence for the mean direction of the sites. Open circles, north-seeking directions down. Solid circles, north-seeking directions up. † = present earth's field, north-seeking down.

All the specimens were subjected to alternating field demagnetization by means of an apparatus similar to that described by McElhinny (1966). At least two specimens from each site were demagnetized progressively in fields up to 70 mWb/m^2 (700 gauss). The test specimens from sites 1, 2, 5, and 6 behaved in a very similar fashion. In fields up to 10 to 15 mWb/m^2 a change of direction of magnetization occurred, which varied in magnitude from specimen to specimen but was generally away from the direction of the present earth's field. The directions of magnetization showed little further change after continued demagnetization in fields up to 70 mWb/m^2. It is apparent that a stable component of magnetization was isolated following the initial removal of a viscous component. The rest of the specimens from these sites were then treated in an optimum field for each site, chosen from the results of the initial progressive demagnetization procedures. Details of the mean directions of magnetization after alternating field demagnetization at each of these four sites are presented in Table I.

One test specimen from site 4 showed a very stable end point from 5 to 70 mWb/m^2 whereas the other was stable in fields from 20 to 50 mWb/m^2. The specimens from this site were demagnetized in fields of 20 and 30 mWb/m^2. The mean direction of magnetization of the samples from this site, after demagnetization in these fields, showed little change from the mean NRM direction. However, the value of k, the estimate of Fisher's (1953) precision parameter κ, decreased from its NRM value of 37.4 to 25.6 in 20 mWb/m^2 and 20.4 in 30 mWb/m^2. These decreases are not significant according to the test devised by McElhinny (1964) and as satisfactory 'end points' were obtained during demagnetization of the test specimens, it is believed that the direction of magnetization after demagnetization in 20 mWb/m^2 represents the stable direction of magnetization for this site.

The 'end points' of the directions of magnetization of specimens from site 3 were not as stable as those of the test specimens from the other sites. The remaining specimens from this site were therefore demagnetized in fields of 10, 20, 30, and 40 mWb/m^2. In none of these fields was the grouping of the directions of magnetization improved significantly. However, the mean direction of the samples changed considerably after demagnetization in 10 mWb/m^2 and then remained fairly constant between 10 and 40 mWb/m^2. The mean direction quoted in Table I for this site corresponds to the demagnetizing field within the range 10–40 mWb/m^2 for which the best grouping was observed, as evidenced by the magnitude of k.

The site mean directions after alternating field demagnetization are shown in Fig. 2, and they are closely grouped well away from the direction of the present earth's field at the Michikamau Intrusion.

Discussion

The site mean directions after alternating field demagnetization have been combined to give a mean direction of magnetization for the intrusion. The parameters of this mean direction are N = 6 sites, D = 262.2°, I = +9.6°, k = 34.9, and α_{95} = 9.7°, where the symbols are the same as for Table I. If the site mean directions are first corrected for the appropriate dip of the igneous lamination or layering (by making them horizontal by rotation about the strike direction) then the mean direction for the intrusion changes to D = 268.9° and I = −4.7°, and k decreases to 7.8. Use of the test devised by McElhinny (1964) shows that the observed change in k is significant at the 95% confidence limit. This provides evidence that the observed magnetization was acquired with the rocks in their present attitudes. The stability of the magnetization of the rocks during the alternating magnetic field demagne-

TABLE II

Pole positions for Precambrian rock units of North America

Rock Unit	Age (m.y.)	Lat.	Long.	References
Michikamau anorthosite	1400±80	0.5° S	144.7° W	
Allard Lake anorthosite	uncertain ~1000 ?	38.6° S	219.6° W	Hargraves and Burt (1967)
Molson dikes	1445 ?	36.2° N	108.9° W	Fahrig *et al.* (1965)
Belt Series, Grand Canyon Series, and Purcell System	>1100<1600	12° S	151° W	Irving (1964)
Purcell System	>1100<1600	18° S	147° W	Irving (1964)
St. Francois Mountains, Missouri	1300–1550	1° S	144° W	Hays and Scharon (1966)
Front Range, Colorado–Wyoming	1410±30	8° S	151° W	Eggler and Larson (1968)

tization procedures is believed to indicate that the observed magnetization is thermo-remanent magnetization acquired when the rocks originally cooled.

The angular dispersion $\Delta = 81/k^{\frac{1}{2}}$ with $k = 34.9$, is 13.7°. The use of an F-test in the manner outlined by Gough *et al.* (1964) provides an upper limit for the value of Δ of 23.6° at the 95% confidence level. This is greater than that calculated by Creer (1962) for the present earth's field at low latitudes (approximately 18°). It is possible therefore that the sampling of the Michikamau Intrusion has been sufficiently detailed that the paleosecular variation is fully represented in the measurements. The calculation of a pole position from the mean direction for the intrusion on the assumption of a geocentric dipole field should thus provide the position of the paleomagnetic pole relative to the region sampled. This calculation places a paleomagnetic pole at latitude 0.6° S and longitude 145.6° W with $\delta p = 4.9°$ and $\delta m = 9.8°$, where δp and δm are the semiaxes of the oval of 95% confidence for the mean pole position. Alternatively the paleomagnetic pole position can be obtained by combining the pole positions calculated from individual site mean directions. The result of this calculation is shown in Table II together with pole positions determined for the Precambrian rock units of North America referred to below.

The pole position for the Michikamau Intrusion differs greatly from that calculated by Hargraves and Burt (1967) for the Allard Lake anorthosite intrusion, and also from that given by Fahrig *et al.* (1965) for the Molson dikes. However, the time of magnetization of the former rocks is uncertain and may well lie during the Grenville orogeny. The age reported by Fahrig *et al.* (1965) for the time of intrusion of the Molson dikes (1445 m.y.) is similar to that of the Michikamau Intrusion. However, it is based on a single whole-rock K–Ar determination on a sample of low potassium content and therefore the age equivalence of these dikes and the Michikamau rocks cannot be considered established. In addition the paleomagnetic result from the dikes is based on a very small number of samples.

Irving (1964) has calculated a combined pole position for the Belt Series of Montana and the Grand Canyon Series of Arizona (Collinson and Runcorn 1960) and the Purcell System of British Columbia (Norris and Black 1961), and also a mean pole position for the Purcell System from additional results by Black (1963). These both lie very close to the pole position determined from the Michikamau rocks. In addition Runcorn (1964) has reported further paleomagnetic results from rocks from the Belt Series and other Precambrian formations from the western United States. The pole positions deduced from the various formations studied by Runcorn (1964) (not recorded in Table II) again lie in the same general region as the Michikamau pole position. The exact age of the rocks from western Canada and the western United States are however uncertain, but may lie within the limits 1100 m.y. (Runcorn 1964) to 1600 m.y. (Giletti 1966).

Hays and Scharon (1966) have reported measurements on some Precambrian igneous rocks from the St. Francois Mountains of southeast Missouri. A mean pole position has been calculated from their published data giving unit weight to each of the five rock units from

which they obtained results. This calculation provides a pole position almost identical to that of the Michikamau rocks. However, the rock units from which they reported measurements are subject to uncertainties with regard to their absolute and relative ages. These rock units bracket granite, which did not give useful paleomagnetic results, but whose age probably lies within the range 1300 to 1550 m.y. (Tilton *et al.* 1962). The differences in times of intrusion between the granite and the rock units that yielded paleomagnetic results is unknown. The combination of directions of magnetization from the rock units of the St. Francois Mountains is, therefore, of uncertain validity, and the striking similarity of the pole positions from Michikamau and the St. Francois Mountains is of debatable physical significance.

Recently Eggler and Larson (1968) have presented a paleomagnetic pole position for some igneous Precambrian rocks from the Front Range of Colorado–Wyoming, whose age has been precisely determined by the Rb–Sr isochron method as 1410 $\pm$ 30 m.y. The similarity in ages of their rocks and of those reported in this paper is paralleled by the similarity in their respective pole positions. The angular distance between the pole positions is only 10°. The striking correspondence of ages and paleomagnetic pole positions suggests that there have been no major relative movements between the sampled regions, separated by several thousand kilometers, since these rocks were formed.

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